

HITIKA HEMNANI

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EDUCATION

University of Southern California, Los Angeles, California	Aug 2024 - May 2026
Master of Science in Computer Science	(GPA 3.73/4)
Relevant coursework: Analysis of Algorithms, Database Systems, Information Retrieval and Web Search Engines	
Vivekanand Education Society's Institute of Technology, Mumbai, India	Jun 2018 - May 2022
Bachelor of Engineering in Computer Science	(CGPA 9.68/10)
Relevant coursework: Data Structures, Object-Oriented Programming, and Databases	

TECHNICAL SKILLS

- Languages: Python, Java, C, SQL, TypeScript
- Backend & Systems: REST APIs, Microservices, Node.js, Django
- Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, CI/CD, Git, GitHub Actions
- Frontend: React.js, HTML, CSS
- Data & ML: TensorFlow, PyTorch, Pandas, NumPy, OpenCV
- Tools: Linux, Debugging, Monitoring

PROFESSIONAL EXPERIENCE

University of Southern California, Los Angeles, CA	Feb 2025 - Dec 2025
<i>Research Assistant (Software & Systems Development) – Department of Computer Science</i>	

Mental Health Project | Python, Docker, React, TensorFlow, CI/CD

- Owned the design, deployment, and operation of a distributed backend pipeline for EEG data processing, enabling near real-time inference and improving end-to-end latency by 22% using containerized services.
- Built and maintained distributed training workflows with automated preprocessing, debugging failures via logs and metrics, reducing manual configuration time by 30%.
- Developed a React-based dashboard for visualizing EEG inference results in real time.

LTI Mindtree, Mumbai, India

Jun 2022 - Mar 2024

Software Engineer – SAP Business Unit

- Engineered scalable backend integration services across 20+ global regions using Python/SAP APIs, reducing incident recurrence by 30% and improving deployment reliability.
- Automated HR workflows and system processes for enterprise clients across multiple geographies, improving system reliability and performance by 27%.
- Collaborated with cross-functional teams to debug and resolve production issues across authentication, networking, and integration layers, ensuring service reliability.

Trivia Software's, Thane, India

May 2021 - Jun 2021

Software Developer Intern

- Developed and deployed a responsive full-stack web application using Django and React/TypeScript, reducing page load time by 40%.
- Integrated REST APIs with frontend state management for dynamic, real-time data updates.
- Implemented PyTest and Jest frameworks for complete functional testing in CI/CD pipelines.

ACADEMIC PROJECTS AND PAPERS

Down Syndrome Detector | React, Django, Python, SQL - [Publication](#)

- Designed an accessible React-based UI integrated with a CNN backend exposed via REST APIs for real-time predictions and patient feedback.
- Published the system in peer-reviewed journals for its innovation in healthcare accessibility.

University Canteen Management System | React, Python, JavaScript, SQL - [Publication](#)

- Developed a full-stack app enabling real-time food orders and personalized recommendations.
- Optimized frontend component rendering and improved checkout time by 35%.

Drone Search and Rescue | Python, AirSim, Unreal Engine

- Implemented autonomous exploration with LiDAR obstacle avoidance and vision-based detection, debugging multi-threaded sensor pipelines to meet real-time constraints.

Transfer Learning for Waste Image Classification | Python, Keras, TensorFlow, OpenCV - [GitHub](#)

- Implemented and compared VGG16, ResNet, and EfficientNet; achieved AUC 0.96 with VGG16.
- Performed error analysis and hypothesis testing to evaluate performance gains.

EXTRACURRICULAR ACTIVITIES

Student Worker, USC Auxiliary Services

Automated event booking workflows and reporting tools using Python/SQL, reducing manual reconciliation by 40%.

Research Publications

Published two peer-reviewed papers in Springer and IJCRT on AI in healthcare and smart systems.

Technical Achievements

Completed JPMorgan's virtual software engineering program focused on data visualization.